BGP Hijack Simulator

An Interactive Learning Environment for Routing Security



Internship Report

Submitted to

National Internet Exchange of India (NIXI)

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1 Introduction

The Internet serves as the backbone of modern communication, connecting billions of devices across the globe. At the heart of this massive infrastructure lies the Border Gateway Protocol (BGP), a standardized exterior gateway protocol used to exchange routing and reachability information among autonomous systems (ASes). BGP plays a critical role in ensuring that data packets are correctly routed from source to destination, even across disparate and independently managed networks.

Despite its significance, BGP was not designed with robust security mechanisms in mind. Over the years, this has made it vulnerable to a range of attacks, most notably BGP route hijacking. In a BGP hijack, a malicious or misconfigured autonomous system falsely advertises ownership of IP prefixes it does not control, thereby redirecting, intercepting, or blackholing network traffic. Such incidents can have severe consequences, including disruption of services, traffic interception, and even widespread Internet outages.

While numerous detection tools and mitigation strategies exist, a major gap remains in the area of hands-on education. Understanding how BGP works—and how it can fail—is crucial not only for network operators and cybersecurity professionals but also for students and researchers. However, due to the distributed and abstract nature of BGP, it is often challenging to grasp these concepts without interactive, visual, or experiential tools.

This project was conceptualized during my internship at the National Internet Exchange of India (NIXI) with the objective of developing a BGP Hijack Simulator. The goal is to create an intuitive, interactive, and educational platform where learners can explore the BGP protocol, experiment with hijack scenarios, and visually understand their impact on routing. The simulator is designed to serve as a learning bridge—bringing together theoretical understanding and practical insight—by providing a user-friendly environment for simulating real-world routing anomalies in a controlled setting.

Through this report, I document the motivation, system architecture, implementation details, and educational impact of this simulator. The project aligns with NIXI's broader mission of strengthening India's Internet infrastructure and promoting knowledge dissemination in critical areas such as routing security and cyber hygiene.

2 Objective of the Project

The primary objective of this project is to design and develop an interactive, browser-based BGP Hijack Simulator tailored for educational use. The simulator is intended to help students, researchers, and network enthusiasts better understand the structure of the Internet, how the Border Gateway Protocol (BGP) functions, and how various types of BGP hijacks occur in real-world scenarios.

Core Objectives

1. To create a visual and interactive platform that allows users to simulate BGP routing behavior and hijack scenarios in real time.

2. To provide a guided learning experience where users can experiment with different types of hijacks including origin hijacks, forged AS paths, prepending errors, and typographical mistakes.

3. To demonstrate the effect of hijacks on routing paths using intuitive network graphs and scenario toggles.

4. To bridge the gap between theoretical BGP concepts and their practical consequences in a safe, controlled, and reproducible environment.

5. To promote awareness around routing security and support India's broader vision of strengthening its cyber infrastructure and digital resilience.

Learning Outcomes

- Users will gain hands-on experience with BGP routing concepts and anomalies.
- The simulator will enable easy comprehension of how seemingly minor misconfigurations can have major impacts on global connectivity.
- Users will be able to identify and distinguish between different hijack types through simulated traffic flows and routing changes.

3 Literature Review

The Border Gateway Protocol (BGP) has been a subject of ongoing research for decades due to its foundational role in Internet routing and its vulnerabilities to misconfiguration and malicious attacks. This section reviews prior work that informed the design of this project and provides context on the challenges in BGP security, especially hijack detection and classification.

BGP and Routing Security

BGP, as defined in RFC 4271, lacks built-in security mechanisms to verify the legitimacy of route announcements. Consequently, any Autonomous System (AS) can advertise any IP prefix, intentionally or accidentally. Numerous real-world incidents—such as the 2008 YouTube hijack by Pakistan Telecom and the 2010 China Telecom incident—have highlighted the critical need for improved BGP security mechanisms.

To address these vulnerabilities, mechanisms such as Resource Public Key Infrastructure (RPKI), prefix filtering, and route origin validation have been proposed. While effective in reducing accidental misconfigurations, these solutions have not eliminated the risk of hijacks due to partial adoption and complex deployment.

Existing Detection and Analysis Tools

Several tools have been developed to detect and analyze BGP hijacks:

- **BGPMon**: A real-time monitoring system that issues alerts when suspicious changes in route origin or AS path are observed.
- **BGPStream**: An open-source software framework developed by CAIDA that facilitates historical and real-time BGP data collection and analysis.
- **RIPE RIS and RouteViews**: Data collection platforms that provide global BGP snapshots for research and operational use.

These tools are highly valuable for operators and researchers, but they are generally not beginner-friendly and lack interactive, educational interfaces.

BGP Hijack Classification Research

The 2018 research paper titled *"BGP Route Hijacking Classification"* by Apostolaki et al. presents a comprehensive taxonomy of BGP hijacks. It categorizes hijacks into five main types:

1. Origin Hijacks
2. Forged AS Paths
3. Prepending Mistakes
4. Typographical Errors
5. Multiple Origin AS (MOAS) conflicts

The study also presents a machine learning-based classifier to distinguish these events based on path features such as AS path edit distance, route similarity, and AS hegemony metrics. This classification framework served as a foundational reference for defining the hijack scenarios implemented in this simulator.

Need for Educational Simulators

While the technical literature is rich in detection strategies and datasets, there is a scarcity of tools that allow learners to experiment with BGP hijacks in a low-risk and interactive environment. This gap highlights the need for simulators that can visually and intuitively demonstrate how routing anomalies occur, and how they affect global traffic flows. The simulator developed in this project attempts to fill this gap by offering an accessible learning tool for students, network enthusiasts, and early-career security professionals.

4 System Design and Architecture

The BGP Hijack Simulator has been designed as a lightweight, browser-based platform that enables users to visualize and interact with various BGP routing and hijack scenarios. The system is modular, extensible, and optimized for educational use.

4.1 High-Level Overview

The simulator comprises three major layers:

1. **Frontend Interface**: A user-friendly graphical interface developed using React.js and Cytoscape.js to render autonomous systems (ASes) as interactive nodes and simulate path announcements and hijacks.
2. **Simulation Engine**: JavaScript-based logic that mimics BGP behavior by manipulating static or dynamic JSON datasets representing AS paths and prefix advertisements.
3. **Scenario Framework**: A set of pre-defined BGP routing scenarios (normal, origin hijack, forged path, etc.) stored in structured files and rendered upon user selection.

4.2 Technology Stack

The project uses modern web technologies to ensure ease of access, performance, and maintainability:

- **Frontend Framework:** React.js (via Vite) for fast rendering and modular development.
- **Graph Visualization:** Cytoscape.js for drawing and updating dynamic network topologies.
- **Routing Data:** JSON-based representations of prefixes, ASes, and paths.
- **Package Manager:** NPM for dependency management.
- **Styling:** Tailwind CSS and custom CSS for clean UI design.

4.3 System Components

Graph Visualization Engine

The graph visualizer represents each AS as a node and each BGP relationship or AS path as an edge. Different path types are colored and styled distinctly to help the user understand how routing changes in different hijack scenarios.

Simulation Controller

The control panel allows users to switch between predefined scenarios. When a scenario is selected, the simulator dynamically loads the corresponding AS path and re-renders the graph to reflect route changes.

Routing Scenario Data Model

The simulation reads from a static JSON dataset containing:

- A list of ASes and their relationships
- A set of IP prefixes and their original owners
- BGP paths for different scenarios

This approach allows easy addition of new scenarios without modifying the core logic.

UI Feedback Components

Scenario descriptions, current mode indicators, and pop-up hints are used to explain the user's actions and the meaning of each route change. These components make the learning process intuitive and engaging.

5 Implementation

The implementation phase involved developing a fully functional web-based simulation platform using React and Cytoscape.js. The simulator allows users to visualize and interact with multiple BGP hijack scenarios, enabling hands-on learning in a controlled environment.

5.1 Frontend Development

The frontend was developed using the Vite build tool with React as the framework. Cytoscape.js was integrated as the primary graph rendering library to display autonomous systems (ASes) and BGP path relationships.

Key React components include:

- **Graph.jsx** – Renders the network graph based on the selected BGP scenario.
- **Controls.jsx** – Provides scenario-switching buttons (e.g., simulate origin hijack).
- **App.jsx** – Acts as the controller, manages state, loads data, and passes props to child components.

The graph layout uses a breadth-first arrangement, which helps emphasize AS path direction and hierarchy. Each node represents an AS, while edges represent the advertised BGP paths.

5.2 Data Modeling

The routing data is modeled using a single JSON file (bgp_data.json) placed in the public directory. This dataset contains:

- List of ASes with unique IDs and labels
- IP prefixes with their original owners
- Path lists for each scenario (e.g., normal, hijack_origin_change)

This static modeling approach allows for easy modification and extensibility. Users or developers can add more hijack types by simply updating the JSON structure without touching core logic.

5.3 Simulation Logic

The simulator does not attempt to fully implement BGP protocol internals. Instead, it mimics route propagation and selection logic by interpreting predefined paths and dynamically displaying the result.

When a user selects a scenario:

1. The corresponding AS path from bgp_data.json is parsed.
2. Graph nodes and edges are generated dynamically from the path data.
3. Cytoscape renders the graph, highlighting affected routes or anomalies.

No backend or real-time networking logic is used, keeping the application lightweight and frontend-only.

5.4 User Interactions

The application supports direct user interaction through:

- Clickable buttons to simulate different hijack types.
- Visual cues such as colored edges to represent legitimate and malicious paths.
- Labels and hover effects for nodes and edges to clarify AS roles.

Future versions may support drag-and-drop AS creation or custom scenario design by users.

6 Scenarios Implemented

The simulator supports multiple BGP hijack scenarios, each designed to illustrate a distinct type of vulnerability or misconfiguration observed in real-world routing environments. These scenarios are implemented using variations in the AS path structure and are visually rendered to show how traffic flow changes when a hijack occurs.

Each scenario is selected from a control panel and loaded into the simulation environment via predefined path data in `bgp_data.json`. This approach ensures consistency and educational clarity.

6.1 Normal Routing

This is the baseline scenario that displays the legitimate BGP path from an origin Autonomous System (AS) to various transit and customer ASes. It helps learners understand how BGP routes are typically formed and propagated.

Example Path: AS4 → AS3 → AS2 → AS1 (AS1 is the original owner of prefix 192.0.2.0/24)

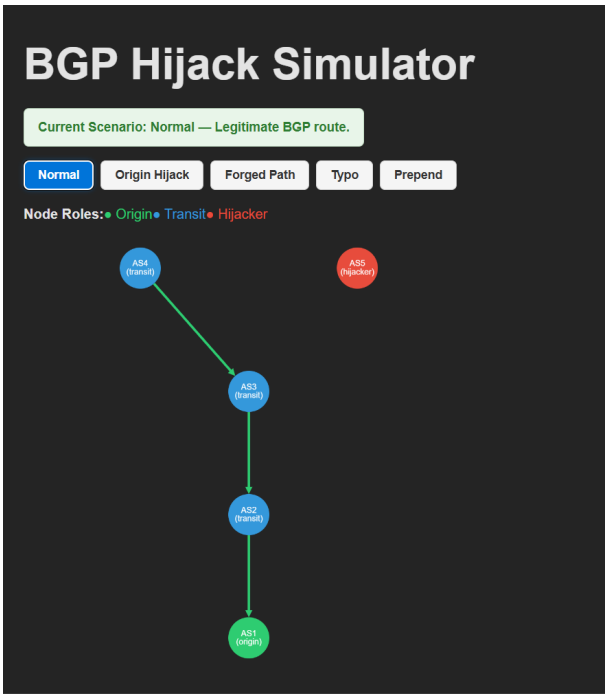


Figure 1: Normal Routing View

6.2 Origin Hijack

An unauthorized AS (e.g., AS5) falsely claims ownership of a prefix it does not control. As a result, some or all Internet traffic may be diverted to the hijacker.

Example Hijacked Path: AS4 → AS3 → AS2 → AS5 (AS5 pretends to be the origin of 192.0.2.0/24)

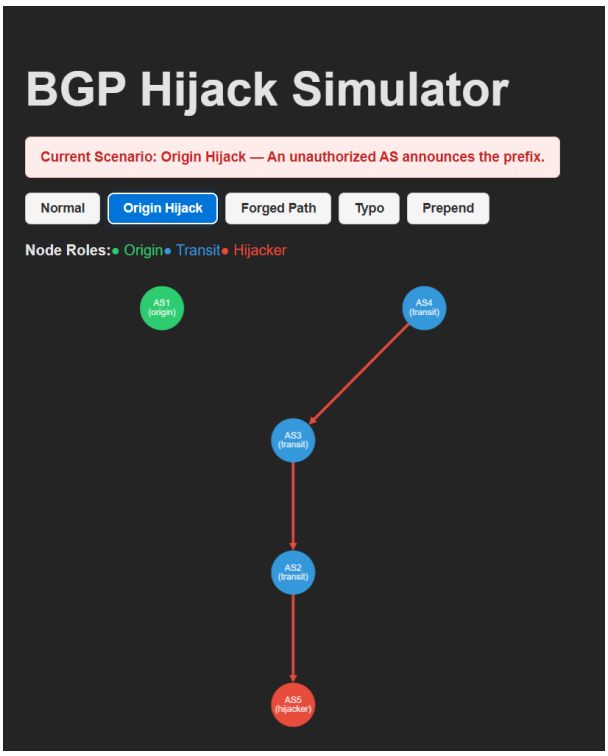


Figure 2: Origin Hijack

6.3 Forged AS Path

The hijacker (AS5) creates a falsified AS path that includes the legitimate origin AS (AS1), tricking others into believing the route is authentic. This form of attack may be harder to detect and can be used for man-in-the-middle attacks.

Example Forged Path: AS4 → AS3 → AS5 → AS1 (AS5 inserts AS1 in the path to increase legitimacy)

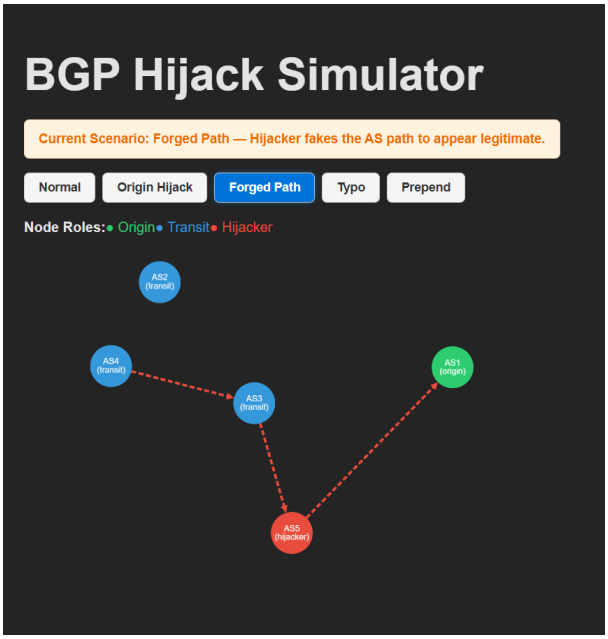


Figure 3: Forged AS Path

6.4 Prepending Error

AS1 attempts to influence route selection by prepending its AS number multiple times (e.g., for traffic engineering), but excessive or incorrect prepending can make routes appear suspicious or less preferred.

Example Path: AS4 → AS3 → AS2 → AS1 → AS1 → AS1

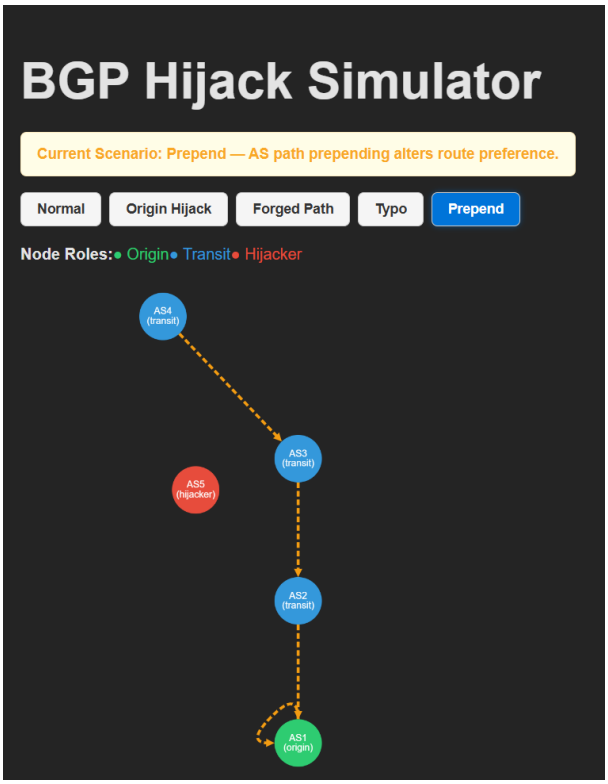


Figure 4: Prepending Error

6.5 Typographical Error (Typo)

An operator accidentally mistypes an AS number or IP prefix, leading to unintended hijacks or routing disruptions.

Example Path: AS4 → AS3 → AS2 → AS2 (AS2 appears twice due to a typo or misconfiguration)

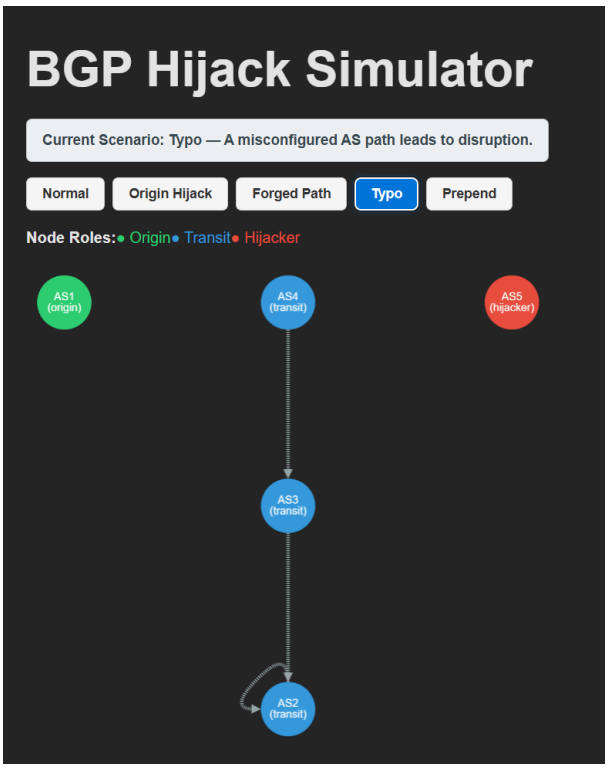


Figure 5: Typographical Error

7 Usage and Demonstration

This section provides a walkthrough of how to use the BGP Hijack Simulator, explains its user interface, and demonstrates the effect of each hijack scenario using simulated data and visual feedback.

7.1 How to Run the Simulator

The simulator is a frontend-only web application. It can be run locally or hosted online using platforms like GitHub Pages or Vercel. To run locally:

1. Ensure Node.js and npm are installed.
2. Clone the project repository:

```
git clone https://github.com/aapJoysakoi/BGP.git
cd BGP
```

3. Install dependencies:

```
npm install
```

4. Start the development server:

```
npm run dev
```

5. Open the browser and go to:

```
http://localhost:5173/
```

7.2 Walkthrough

The main screen displays:

- A graph visualization of ASes and paths using Cytoscape.js.
- A control panel with buttons for selecting routing scenarios.
- A display showing the currently active hijack type.

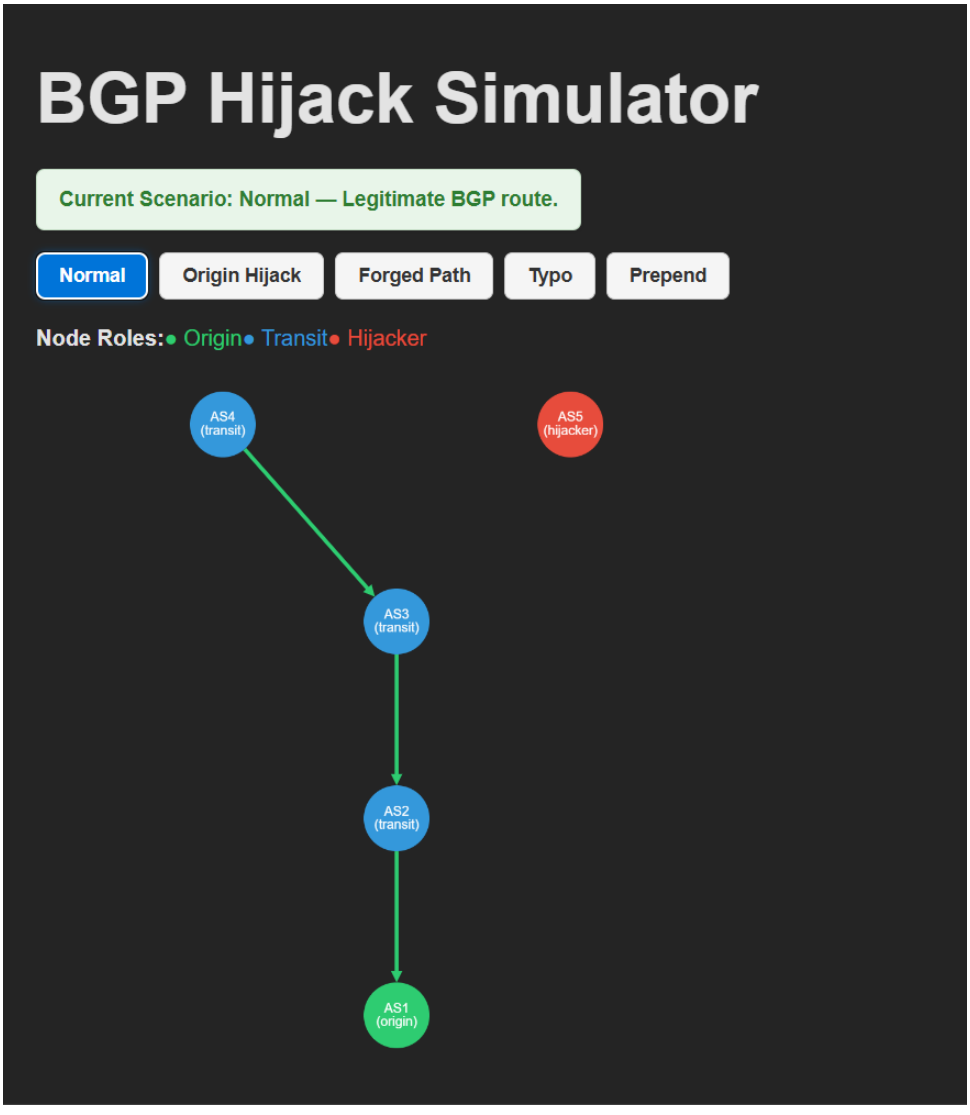


Figure 6: Simulator Interface: Normal Routing View

Scenario Buttons:

- Normal
- Origin Hijack
- Forged Path
- Typo
- Prepend

Each button dynamically updates the graph to show the corresponding BGP state.

7.3 Sample Simulation Walkthrough

To simulate a hijack:

1. Click on the "Origin Hijack" button.
2. Observe how the original path (e.g., AS1 → AS2 → AS3 → AS4) changes.
3. The new path now ends at AS5, representing the hijacker.
4. Hovering over nodes and edges shows AS labels and relationships.

Visual cues such as edge colors and labels help users differentiate between normal and hijacked paths.

8 Learnings and Challenges

The development of the BGP Hijack Simulator provided an opportunity to gain in-depth technical knowledge, hands-on experience with modern frontend technologies, and a deeper appreciation for Internet routing security challenges. This section reflects on the key learnings and the challenges faced during the course of the project.

8.1 Technical Learnings

- **Understanding BGP Fundamentals:** Developing the simulator required a detailed understanding of how BGP functions at the AS level, including route advertisement, AS path selection, and prefix ownership.
- **Types of BGP Hijacks:** Gained clarity on the various types of hijacks such as origin hijacks, forged AS paths, prepending anomalies, and MOAS conflicts. Learning how each affects route propagation helped in modeling accurate simulations.
- **Graph Rendering with Cytoscape.js:** Learned how to dynamically generate and update network graphs using Cytoscape.js, which involved programmatically building nodes, edges, and layouts based on route data.
- **Frontend Architecture using React:** The project deepened understanding of React component structures, props, state management, and how to build modular UIs efficiently using tools like Vite.
- **Data Modeling with JSON:** Structured mock BGP data to represent multiple scenarios and built reusable logic to feed this data into the graph rendering engine.

8.2 Cybersecurity and Routing Concepts

- The project helped connect cybersecurity theory with networking practice. BGP vulnerabilities that were once abstract became concrete through visualization and simulation.
- Understood the risks associated with non-secure routing protocols and the limited deployment of protection mechanisms like RPKI and BGPsec.
- Gained awareness of real-world incidents caused by BGP hijacks, and how misconfigurations can propagate globally if left unchecked.

8.3 Major Challenges and Solutions

- **Modeling BGP Behavior Without a Real Protocol Stack:** One of the core challenges was simulating BGP behavior without implementing the full protocol. This was addressed by using static path definitions and focusing on visual flow rather than protocol logic.
- **Designing for Clarity and Simplicity:** It was important to ensure that the simulator remained intuitive and educational. Striking a balance between technical depth and ease of use required iterative design and user feedback.
- **Learning Cytoscape API and Styling:** Initially, styling the graph and managing dynamic layouts was complex. This was resolved through experimentation, reading Cytoscape documentation, and breaking the UI into components.
- **Scenario Switching Logic:** Managing consistent updates to the graph when switching between multiple hijack types required thoughtful state management and re-render logic in React.

9 Future Scope

While the current version of the BGP Hijack Simulator provides a strong foundation for visual and interactive learning, there are several opportunities to expand its features and utility in future iterations. This section outlines potential improvements and directions for scaling the project further.

9.1 CLI-based Learning Mode

Introducing a command-line interface (CLI) version of the simulator could enhance accessibility for users who prefer terminal-based environments. This would also allow integration with scripting and more advanced automation workflows. Users could issue commands like:

```
announce 192.0.2.0/24 from AS1
simulate hijack origin 192.0.2.0/24 by AS5
show routing-table
```

9.2 Realistic Routing Lab with FRRouting and Docker

Incorporating a containerized lab setup using tools like FRRouting, BIRD, and Docker or Mininet would provide real-world simulation capabilities. This would be especially useful for advanced learners or cybersecurity professionals who want to experience actual BGP behavior, convergence, and hijack detection.

- Each container could represent an AS running real BGP software.
- Hijacks could be triggered by changing route configurations in isolated testbeds.

9.3 Scenario Builder and User-Created Simulations

Enabling users to build their own routing topologies and hijack scenarios through a drag-and-drop interface would significantly improve engagement. This feature could allow learners to create and save custom topologies, set prefix origins, and simulate arbitrary AS paths.

9.4 Gamified Learning Experience

Gamifying the simulator could improve knowledge retention and user engagement. Examples include:

- Challenge mode: "Detect the hijack in 3 scenarios"
- Score-based system based on correct identification and mitigation
- Time-limited response modes to mimic real-world alert handling

9.5 Integration with Real BGP Datasets

The simulator could be enhanced to allow playback of real BGP anomalies from sources such as RouteViews, RIPE RIS, or BGPStream. This would enable users to analyze actual incidents and compare them with simulated versions.

9.6 Educational Curriculum Integration

The project could be packaged with modular lesson plans and interactive labs for integration into university-level cybersecurity or networking courses.

The BGP Hijack Simulator project represents a meaningful step toward making complex Internet routing concepts more accessible, especially in the context of cybersecurity and network governance. Developed during an internship at the National Internet Exchange of India (NIXI), this project bridges the gap between theoretical knowledge and hands-on experience by offering a visual, interactive platform for learning about BGP and its vulnerabilities.

Through the simulator, users are able to explore how BGP operates under normal conditions and how various hijack techniques—such as origin forgery, AS path manipulation, and configuration errors—can lead to significant routing disruptions. Each scenario is carefully designed to prioritize educational clarity and user engagement, making the tool suitable for students, researchers, and early-career professionals.

The frontend-only architecture ensures lightweight deployment and easy scalability, while the modular data-driven design enables new scenarios to be added with minimal effort. With future enhancements such as CLI mode, containerized real-world labs, and gamification, the simulator has the potential to become a widely used learning tool in the field of routing security.

This project not only reinforced technical skills in React, Cytoscape, and data modeling but also deepened my understanding of critical Internet infrastructure and the risks posed by insecure routing. It aligns closely with NIXI's vision of strengthening India's digital ecosystem and fostering cybersecurity awareness across the Internet community.

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